

BUIDL CTC 2026 FALL · DEFI

Kitty

Savings circles where every payment is proven, not promised.

Ethereum stablecoins · settled and credit-scored on Creditcoin · via the Attestcoin Protocol

PROBLEM

Hundreds of millions save in circles nobody can see.

- Chit funds, susu, tandas, chamas, stokvels: ten friends, one pot a month.
- They fail two ways: the treasurer runs, or a member stops paying and nobody outside the group ever knows.
- Ten years of perfect payments builds zero formal credit history.

TRUST TODAY

Who you have to trust

STEP	INFORMAL CIRCLE	“ROSCA ON-CHAIN” APPS	KITTY
Holds the money	Treasurer	One contract on one chain	Escrow vault on Ethereum
Knows who paid	Treasurer’s notebook	Same contract	Attestcoin-proven txs on Creditcoin
Decides the deadline	Treasurer	block.timestamp / admin	Attested source-chain block height
Picks the recipient	Treasurer / lottery	VRF oracle	Deterministic rotation
Credit history	None	None	Kitty Score, readable by any lender

SOLUTION

Money on Ethereum. Rules on Creditcoin. Proofs in between.

- Members pay into a minimal escrow vault on Sepolia.
- The attester network attests the block; one batch proof covers the whole round.
- KittyLedger verifies through precompile 0x0FD2, decodes and binds every payment, enforces deadlines from 0x0FD3.
- Payouts are proven back. “Paid” is never an operator’s word.
- Any member can prove a round from the browser: the Proof Builder serves CORS, the ledger doesn’t care who submits.

LIVE LOOP

**Pay → attest → prove → verify
→ close → pay out → prove back**

Demo: three members, 100 tUSD, round 0 everyone pays; round 1 one member misses; the deadline block is attested; the round closes with a missed record; the score drops from 515 to 395.

What the ledger actually checks

- Batch `verifyAndEmit`: up to 10 txs, one continuity proof, one call per round.
- EvmV1Decoder: receipt status 1, exactly one Contributed log, emitter = registered vault, tx.to = vault, tx.from = member, exact amount, current round.
- Query id = `keccak(chainKey || height || txIndex)`, same as ASCBase; chain key pinned.
- Deadlines from attested height:
`is_height_attested(chainKey, deadline)`.
- Proven against the live precompile before deploying: a real Sepolia proof $\rightarrow$ `0x0FD2.verify = true`; tampered bytes and a wrong chain key revert (`pnpm verify:live`).

ATTACK LAB

Five ways to cheat, five decoded rejections

- Replay → QueryAlreadyProcessed
- Fake vault with a perfect event → WrongEmitter
- Same proof, chain key 3 → WrongChain
- Included but reverted tx → SourceTxFailed
- Paid after the deadline → accepted, flagged late, -20

Reviewed adversarially before submission: only owner-trusted vaults can feed the ledger, nobody can be penalised for a circle they never consented to, a 64-block grace window protects on-time payers from a hostile early close, and pots only go to members who paid this round.

KITTY SCORE → CREDIT

The score is used, not just displayed

- 500 base · +15 on time · −20 late · −120 missed · 300–850.
- KittyCreditLine on Creditcoin underwrites from the score alone: tier A borrows 100% of proven volume, B 50%, C 20%, D nothing.
- A soulbound Kitty Score badge renders live from the ledger, so a lender can read it on any explorer.
- Every input is a proven transaction or an attested deadline. This is the data Creditcoin was built to carry.

From circles to credit lines

- Score-gated circle sizes and seat bidding.
- ERC-5192 Kitty Score badge; lender integrations on Creditcoin.
- Attestcoin writability for payouts once audited; Ethereum mainnet chainKey on CC3 mainnet.
- Pot cover through proven-event insurance.

TEAM

Prashant • solo builder

Foundry • TypeScript • React. Repo: <https://github.com/Prashant-thakur77/Kitty>. Contracts on Sepolia and Creditcoin CC3 Testnet; 27+ Foundry tests; local two-anvil rehearsal; live dashboard.